

### Problem 1

e) Proposal: By combining all three compression methods: pruning, knowledge distillation, and quantization, we are able to minimize the model size while maintaining accuracy. The previous methods each by themselves do decrease the model size, but if we're looking for the most minimization possible, we need to apply all three methods.

By using all three methods, I was able to decrease model size to 5.84kb with a test accuracy of 94.4%. However, the int8 quantized model from before was 5.76kb with an accuracy of 1, a perfect accuracy, and smaller size. This is because the data being linearly separable means a perfect score is possible. However, knowledge distillation and pruning removes connections that are apparently necessary to achieve 100% accuracy. Since the original int8 model is smaller and more accurate, we are not able to improve size or performance with my proposed model.

### Problem 2

1) Types:

- a) Classification: DNN for classifying IMU data into physical activities
- b) Transfer learning: fine tuned to recognize custom items with minimal training data, taken originally from large data sets.
- c) Object detection: Keep track of how many swords are in a frame.
- d) Regression: Turn metrics such as age, weight, and amount of years fencing into prediction of how many rounds someone will win fencing.

- e) Anomaly detection: detect if a motion is a typical fencing motion. If values start to look off, the sword may not be currently being used for fencing, and is instead in transit or something else.
- f) Visual anomaly detection: Trained on images of clean swords, raises an anomaly if it sees a sword that is dirty or otherwise not in mint condition.

## 2) Types:

- a) Dense: Fully connected layer of neurons and weights.
- b) 1D convolution / pooling: Used for features that require spatial information along one dimension, i.e. a simple time-series of sensor values.
- c) 2D convolution / pooling: Spatial information along two dimensions, i.e. images.
- d) Reshape: Turns 1D data into higher dimensional data.
- e) Flatten: Turns higher dimensional data into 1D data.
- f) Dropout: Randomly cuts a portion of network connections during training, good for reducing chance of overfitting.

3) There is unoptimized (float32), and int8 quantized.

4) It supports image and audio modalities for synthetic data generation.

Image generation methods: Black Forest Labs Flux-Pro Synthetic Image Generator,  
DALL-E 3 Synthetic image generator

Voice generator uses OpenAI's text-to-speech model to generate synthetic voice samples.

Synthetic data generation is important because real world training data isn't always easy to come by. With synthetic generation, you can get a much larger volume of data for cheaper and easier than it would be to collect. Also, with synthetic data you can target certain corner cases or things you want to test, whereas with real training data you may not get all of the corner cases you want to test, or maybe not in the quantity you want to test it.